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Electronic nose coupled with artificial neural network for classifying of coffee roasting profile

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ABSTRACT

Coffee known for its diverse aromas shaped by postharvest treatments, particularly the roasting process, plays a pivotal role in determining the quality of the brewed beverage. This study focuses on classifying the aroma of Arabica coffee beans based on roasting temperature, employing an electronic nose equipped with a TGS gas array sensor. The classification methodology integrates deep learning through an artificial neural network (ANN), along with a calculation analysis utilizing the Pearson correlation coefficient. Raw Robusta coffee beans were subjected to five distinct roasting treatments (185 °C, 195 °C, 205 °C, 215 °C, and 225 °C), resulting in light roasts, light to medium roasts, medium to dark roasts, medium to dark roasts, and dark roasts. The repeatability test affirms the TGS sensor's reliability, exhibiting a standard deviation (STD) below 20%. Notably, the TGS 2612 and TGS 2611 sensors, dedicated to odor detection, demonstrated excellent validity with an STD below 10% across various roasting temperatures. Classification results from deep learning cross-validation showcase impressive accuracy: 98.2% for Light Roasts, 98.4% for Light to Medium Roasts, 98.8% for Medium Roasts, 97.8% for Medium Roasts, and 95.9% for Dark Roasts. In conclusion, this study reveals that the *E*-nose, utilizing the TGS gas sensor array with deep learning analysis, effectively detects and classifies coffee types based on roasting time with high accuracy.

1. Introduction

The most famous and expensive thing grown on plantations is coffee, which is consumed in more than 10 million cups every year. This coffee's flavor, background, and custom all contribute to its allure. Several varieties of coffee, including Arabica, Robusta, or a mix of these two, are popular on a worldwide scale. According to a report by an international coffee group, the trade in Robusta coffee is relatively lower than Arabica coffee [1]. When compared to Robusta, which had 47.62 million bags exported in 2021, Arabica had a total of 82.72 million bags. At the end of 2021, the trade in Arabica coffee had grown by 4.1%, and the future looks promising. Arabica is promising both in terms of commerce overall and price. Compared to Robusta, Arabica coffee is generally more costly. The price differential per kg ranges from 1.26 to 2.35 USD [2].

Robusta is cultivated at lower elevations and is renowned for its strong character, while Arabica is valued for its delicate taste and subtle texture. Because Arabica is of more excellent quality and commands higher prices, it dominates worldwide trade. Market impact comes from Brazil, a significant manufacturer [3]. The sector adapts to changing customer needs by incorporating ethical and sustainable practices. Distinctive features, consumer inclinations, and ethical and economic considerations all impact the dynamics of the world coffee market.

A crucial stage in making coffee is roasting, which involves giving green coffee beans heat. The coffee is given a nice aroma by roasting, which also removes the water from the beans. Because roasting changes the chemistry, Arabica coffee has a more complex chemistry, as well as a

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better smell, taste, and color. This makes the roasting process crucial [4]. This procedure, done at different temperatures, produces different roast levels, giving the coffee its particular smells and scents. The beans' chemical transformations produce a range of tastes, including Maillard browning and caramelization, from light to dark roasts. Dark roasts provide more profound, smokier overtones, while light roasts maintain the natural bean qualities. To reconcile maintaining the natural characteristics of the bean with reaching the appropriate roast degree, roasting calls for dexterity and skill. Roasters of coffee manipulate temperature and time precisely to create a harmoniously balanced taste profile. The quality of coffee is greatly influenced by roasting. For the coffee to taste harmonic and pleasurable, careful control over the process is necessary to prevent under- or over-roasting. *Coffee roasting* is a painstaking art form that combines scientific knowledge with sensory perception to transform raw beans into a fragrant, tasty beverage ready to be brewed. Roasting temperature and time determine roasting levels, requiring a roaster to be close by to be examined the complexity of this process accounts for the coffee roaster's expensive compensation. The caliber of specialists is determined by their business. According to an official survey conducted by a third party the average salary for a coffee roaster is between 15,000 and 20.000 USD annually, with the highest salary surpassing 100.000 USD [5]. We are trying to come up with an alternative to forecasting coffee roasting levels based on these expense and complexity issues.

The Electronic Nose (*E*-Nose) is a revolutionary solution to the problems with traditional coffee roasting level assessment techniques. Conventional methods are expensive and complicated since they depend on expert roasters and subjective human judgments. With deep learning and gas sensors, the *E*-Nose provides a reliable and effective substitute. It delivers sensitive, real-time analysis by imitating the nostrils of mammals, addressing problems such as age-dependent sensory fluctuations and subjective "cup tests." The research shows how well the *E*-Nose categorizes Arabica coffee beans according to roasting temperature, providing a consistent and trustworthy method for assessing coffee quality.

Therefore, to address things objectively, technology such as the Electronic Nose (*E*-Nose) has been developed. E-nose has been widely used for quality detection of food ingredients [6] and microbes that cause infection [7] and contaminants [8]. E-nose is a device that mimics the workings of the mammalian nostrils by using gas sensors that can respond to certain scents [9]. The signal response generated by the e-nose to certain scents will be analyzed using pattern recognition software so that it can be analyzed and identified [10]. When compared with other analytical techniques, such as gas chromatography and electronic nose systems can be built in and can provide sensitive and selective analysis in real time. The e-nose system has four main components, namely the gas sensor array, headspace system, data acquisition and pattern recognition [11] [12]. The gas sensors used by *E*-Noses are made of conductive polymer gas sensors, quartz microbalances, surface acoustic vibrations, and metal oxides [13]. The sensing and purging mechanisms make up the two main components of the headspace system. This invention shows that it is possible to effectively convert the chemical reaction between volatile substances and gas sensors into a digital signal [14]. Various studies have used comparisons of various methods to see discrimination during coffee washing [15], as well as coffee recognition using *E*-nose [16] and the use of E-nose and E-tongue to differentiate coffee in China based on species [17].

The roasting degree can be changed to suit customer preferences [18]. Five roasting degrees were used, this research roasting level is not indicative of the real roasting level [19]. Since level five requirements are the most important in this study, clients with lower-level needs can still use our method.

Coffee is classified using multispectral and random forest methods [20]. Linear Discriminant Analysis (LDA) is used to ascertain the type of coffee's flavor [14]. However, all characteristics and predictors must be assumed to have normal distributions in order to use LDA, using a Support Vector Machine (SVM) and Random Forest (RF) [21]. However, it is very challenging to separate data with small vector dimensions due to the spread of data that can be obtained at a particular location. Backpropagation analysis is used to forecast Robusta and Arabica coffee grinds [22]. Unfortunately, extensive study is required to determine backpropagation's ideal performance. Principal Component Analysis (PCA), is a technique related to Gas Chromatography-Mass Spectrometry, which is used to analyze the flavor of Java and Sumatra Robusta coffee [23]. PCA can be used with small data points, but requires a deep learning framework to take full advantage of data.

Six gas sensors (TGS 2620, TGS 2612, TGS 2611, TGS 2602, TGS 2600, and TGS 826) are used to examine the scents produced by Arabica coffee beans at different roasting temperatures. Every sensor provides distinct information. Organic solvents are detected by TGS 2620, combustion gases are targeted by TGS 2612, methane is detected by TGS 2611, and volatile organic chemicals are identified by TGS 2602, cooking. Tobacco scents are targeted by TGS 2600, and ammonia and amines are detected by TGS 826. Combined, these sensors provide a thorough grasp of coffee odors' subtleties and chemical makeup. This is the foundation for a deep learning model that evaluates coffee quality using fragrance profiles.

The model has difficulty dealing with dark roast coffee because of its natural qualities and consistent and robust fragrance profile. Difficulty in distinguishing fragrance qualities emerges from complicated chemical reactions caused by the lengthy, high-temperature roasting process. Sensitivity issues with strong dark roast scents might cause the *E*-Nose system's sensors to miss more minor differences. Subjectivity in human judgments and limited variety in the training dataset adds to the challenge. Accurate categorization is further complicated by the possible overlap in fragrance qualities between dark roast and neighboring roast degrees. A sophisticated strategy that includes feature inclusion, sensor improvement, and dataset refining is needed to overcome these obstacles for better dark roast categorization.

The study uses an electronic nose (*E*-Nose) with TGS gas sensors to forecast coffee roasting levels, specifically Arabica coffee. The *E*-Nose mimics mammalian nostrils and uses pattern recognition software for analysis. The technology offers objectivity and is suitable for quality detection in food ingredients [24]. The E-Nose classifies Arabica coffee beans based on roasting temperature, determining final product characteristics. The deep learning classification achieves high accuracy values, making it a promising alternative to traditional methods for assessing coffee quality based on aroma profiles.

The *E*-Nose sensor is employed to classify Arabica coffee beans based on roasting temperature and to identify the most informative sensor for distinguishing coffee profiles. Raw Robusta coffee beans were divided into five samples, with each sample undergoing a different roasting treatment: roasting at 185 °C, 195 °C, 205 °C, 215 °C, and 225 °C. This process aimed to obtain coffee bean samples representing light roasts, light to medium roasts, medium to dark roasts, and dark roasts.

2. Materials and methods

2.1. Sample preparation

The sample is 2 kg of Robusta coffee beans from the Ijen area, East Java, Indonesia which are still green in color and contain a moisture content of 12.9%. Robusta coffee beans that were still raw were divided into five samples with each sample going through a different roasting process [25], namely roasting at 185 °C, 195 °C, 205 °C, 215 °C, and 225 °C, in order to obtain samples of light roasted coffee beans (S1), light to medium roasted coffee beans (S2), medium to dark roasted coffee beans (S3), medium to dark roasted coffee beans (S4), and dark roasted coffee beans (S5).

Lightly roasted Robusta coffee is produced by a mild roasting method at 185 degrees Celsius. The moderate-to-medium roasting process is carried out at 195 °C. Roasting takes place at 205 °C for the medium roast variety. Roasting takes place at a temperature of 215 °C for the medium to dark roast. Moreover, it is roasted to 225 °C during the dark roasting stage. Data was gathered in the form of light roast, light-to-medium roast, medium roast, medium-to-dark roast, and dark roast coffee beans for each treatment sample, which analyzed with the *E*-Nose up to 50 times.

2.2. Set-up of experiment

The experimental setting for this study is shown in Fig. 1 with an array of gas sensors that collect Robusta coffee aromas during the data collection phase. Previously, pre-heating or heating was carried out on the sensor for 30 min, aiming to stabilize all sensors so that the sensor can run well when it detects samples. Next, the sensor response was tested with 50 replications. The output data obtained by the gas sensor array is the value of voltage over time. Meanwhile, the input value is in the form of the aroma obtained from the sample. During the pre-processing phase, an array of sensors is used to determine the importance of each sensor. Computational analysis was used to predict coffee groups using Pearson's correlation coefficients. (See Fig. 2.)

Six sensors TGS 2620, TGS 2612, TGS 2611, TGS 2602, TGS 2600, and TGS 826 are used in this investigation. Each sensor has a particular gas detection range [20]. More specifically, the TGS 2620 detects vapors of organic solvents and alcohol gases between 50 and 5000 ppm EtOH. Methane, propane, and butane each have a detection range of 1.25% LEL for the TGS 2612. The TGS 2611 can detect methane at levels between 500 and 10,000 ppm. TGS 2602 has a 1 to 30 ppm EtOH detection range for volatile organic substances (hydrogen sulfide, VOC, ammonia, LPG, butane and propane). The TGS 2600 can detect tobacco and cooking odor gas at concentrations between 1 and 30 ppm. The TGS 826 can detect ammonia and amine compound at concentrations between 30 and 300 ppm. Samples were observed using these sensors.

2.3. Pearson correlation analysis

Pearson correlation analysis is often used to choose which features to use or to figure out how to change new features [26,27]. Pearson correlation analysis helps to avoid overfitting and reduce computer work. The Pearson correlation coefficient is usually used to measure the linear relationship between two variables. Its values range from −1 to 1. A score of −1 indicates a perfectly negative correlation. A number of 1 indicates a fully positive correlation, while a value of 0 indicates a linear relationship between two variables. The Pearson correlation coefficient is described as follows:

$$R = \frac{\sum_{i=1}^{n}(x_i - \overline{x})(y_i - \overline{y})}{\sqrt{\sum_{i=1}^{n}(x_i - \overline{x})^2}\sqrt{\sum_{i=1}^{n}(y_i - \overline{y})^2}} \tag{1}$$

where x_i and y_i stand for the X and Y values of the *it* sample, $\overline{x}$ and $\overline{y}$ are the mean values of X and Y, respectively, and n is the number of samples [24].

2.4. Deep learning framework

Gathering information from *E*-Nose sensors when roasting Robusta coffee beans is part of integrating a deep learning framework after Pearson correlation analysis. Critical sensors are identified by correlation analysis, and a double-hidden-layer backpropagation neural network is built to forecast roasting levels. After training and parameter adjustment, the model outperforms conventional techniques regarding accuracy. The method advances the evaluation of coffee quality by offering an efficient and objective way to predict coffee roasting levels based on fragrance patterns.

Neural networks can be used to model and predict nonlinear systems, mimicking any continuous nonlinear map. In this paper, a double-hidden-layer backpropagation neural network with several inputs and six outputs is built. The input system prediction parameters are thought to be representations of the neurons found in the input layer. To complete the spatially weighted aggregate of the excitation input and output signals, the first hidden layer's neuron nodes are employed. Neuron nodes in the second hidden layer are used to enhance the network's nonlinear mapping capabilities for intricate input-output interactions. Two process neuron nodes make up the output layer, which completes the system output.

The transfer function of each layer has a big effect on how well a backpropagation neural network works. Through experimentation, several transfer functions were discovered. The linear transfer function, tangent transfer function, logarithmic sigmoid transfer function, and others are frequently used transfer functions in backpropagation neural networks. Tensor Flow library defines three types of transfer functions: tangents, linear functions, and tangents56. This can be phrased as follows, under the presumption that the signal from the input layer to the first hidden layer is:

$$m_j = \sum_{k=1}^{m} x_k + w_{kj} + b_j \tag{2}$$

where x_k is the input neuron and x_k is each design parameter of the Odor Sensor discussed in this work. w_{kj} represents the input layer to the weight of the first hidden layer, whereas b_j indicates the bias of the first hidden layer. The output signal for the first hidden layer is indicated by y_j in Eq. (3):

$$y_j = \text{tansig}(m_j) \tag{3}$$

The signal transferred from the first hidden layer to the second hidden layer is symbolized by the symbol n_i, and its calculation is depicted in Eq. (4):

$$n_i = \sum_{j=1}^{k} y_j w_{ji} + b_i \tag{4}$$

where w_{ji} is the weight of the second hidden layer's first hidden layer and b_i is the bias of the second hidden layer. The output signal of the second hidden layer is symbolized by the symbol z_i and is expressed as

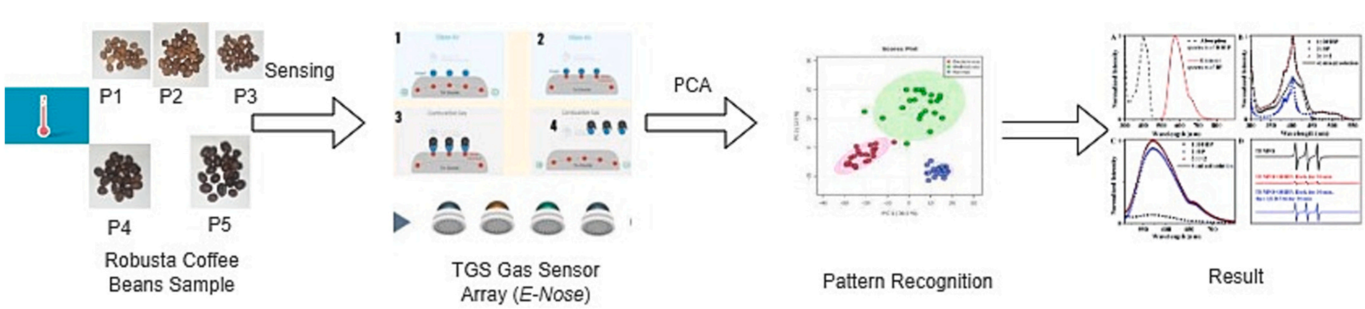

Fig. 1. Experiment setup.

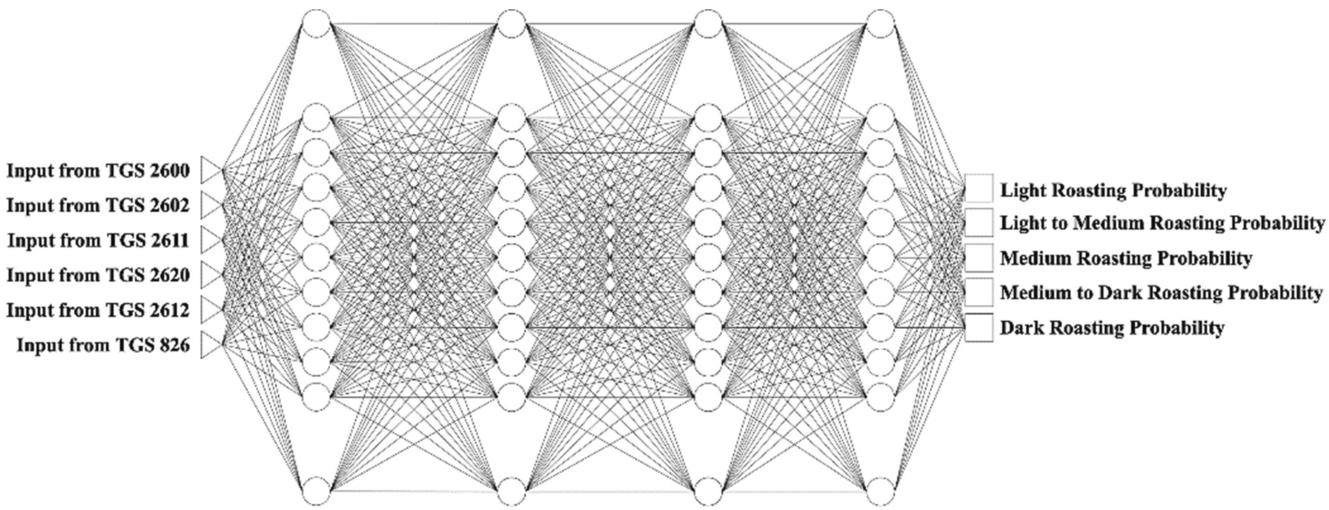

Fig. 2. Deep learning framework.

follows:

$$z_i = \text{tansig}\ (n_i) \tag{5}$$

The signals transferred from the second hidden layer to the output layer are recorded as s, respectively, as illustrated in Eq. (6):

$$z_i = \sum_{i=1}^{p} z_i w_i + b_i \tag{6}$$

2.5. Fine-tuning deep learning procedure

Depending on the amount of data needed, deep learning is often trained using a variety of approaches. Various deep learning techniques often result in different results. The following details our deep learning weighting strategy: Initializing a model in step 1 is rendered absolutely unnecessary. A model needs to be adjusted for a certain objective in order to solve the issue. The starting node of a model typically includes a random node weight and bias. These weights typically vary from 0 to 1. Depending on how challenging the problem is, deep learning uses a different amount of weight.

Forward propagation is the process of performing calculations on data to identify the current model's anticipated label. Step three is to determine the loss function. To determine the current model's performance, a loss function computation is necessary. Generally, the mean square error (data ratio), category cross entropy (ordinal), and other comparable metrics serve as the foundation for the loss function. We use categorical cross-entropy as a loss function because our dataset is ordinal. We have classes and data, where the projected label and the actual label of the data are both present. The following is a definition of the total loss function:

$$L(\widehat{y}_i, y_i) = -\sum_{i}^{N} y_i log \widehat{y}_i \tag{7}$$

Fourth step is Optimization. Optimization is a technique that lowers the overall inaccuracy of the previous step. There are several optimization techniques, including Adam, Adagrad, RMS Prop, etc.;

The Fifth step is back propagation. The backpropagation algorithm reduces the error value in the loss function using the delta rule or gradient descent to obtain the ideal weight. Based on the loss function, this phase decides whether weight needs to be added or subtracted. There are weight modifications from the last layer to the input. Increasing both the error rate and layer weight if the derivative value is positive. The weight must be decreased. If not, layer weight increases. The derivative score of the preceding layer is then calculated.

$$w_n^m = w_n^{m-1} - \epsilon^* \alpha \tag{8}$$

w_n^m is weight in node n and iteration m. ϵ is derivative score and α is learning rate.

2.6. Data collection

For potential future analysis, our dataset is freely available for download on GitHub [1]. Six sensors' worth of raw data, representing classes, total 26,002 bytes in the repository. Fig. 1 demonstrates the normal distribution of the data we have gathered. The dataset has a clustered shape rather than being divided into classes linearly. This is consistent with our belief that natural data will result in a clustered structure. The distribution of our dataset is balanced as well. 19,124 light roast data, 19,122 light to medium roast data, 19,132 medium roast data, 19,508 medium to dark roast data, and 19,116 dark roast data are all included in the dataset distribution. Any classification model would benefit from using this condition as a data distribution. This dataset's intricacy is its biggest flaw. The data features cross over one another. These conditions are difficult and require additional research to be resolved.

3. Results

3.1. Pearson correlation coefficient

We use Pearson correlation to figure out how important each sensor is [23,25]. This helps us find the right feature and deal with the cost issue. We investigated whether the output of a single predictor could be improved by integrating various olfactory descriptors. We conduct the calculations outlined for the training dataset to get the Pearson correlation coefficient between each pair of odor descriptors. The standard for model fusion is the Pearson correlation coefficient. The Pearson correlation coefficients between the several odor descriptors are displayed in Table 1.

As the output of the regressor, each new pair of sensors is compared to a single description. The six sensors are represented by TGS 2600 to TGS 826. Sensor TGS 2600 and sensor TGS 2620 have a close relationship. Sensor TGS 2602 and sensor TGS 826 have a close relationship. 2611 sensors and 2612 have no high relation to other sensors. Fig. 3 shows how the accuracy of predictions only changes for one or two variables based on the output of the regressor. The most effective sensor for predicting coffee is TGS 2602. The TGS 2600 and TGS 2620 sensors are closely related. Despite having similar Pearson correlation

Table 1
Pearson correlation coefficient for roasting profile coffee.

TGS 2600	TGS 2600	TGS 2602	TGS 2611	TGS 2620	TGS 2612	TGS 826
TGS 2602	0.697230	1				
TGS 2611	0.411828	0.686284	1			
TGS 2620	0.991888	0.735301	0.438835	1		
TGS 2612	0.682535	0.498495	0.243995	0.651970	1	
TGS 826	0.710215	0.972227	0.651203	0.749243	0.538378	1
Class	0.750990	0.428502	0.038299	0.776075	0.440386	0.506036

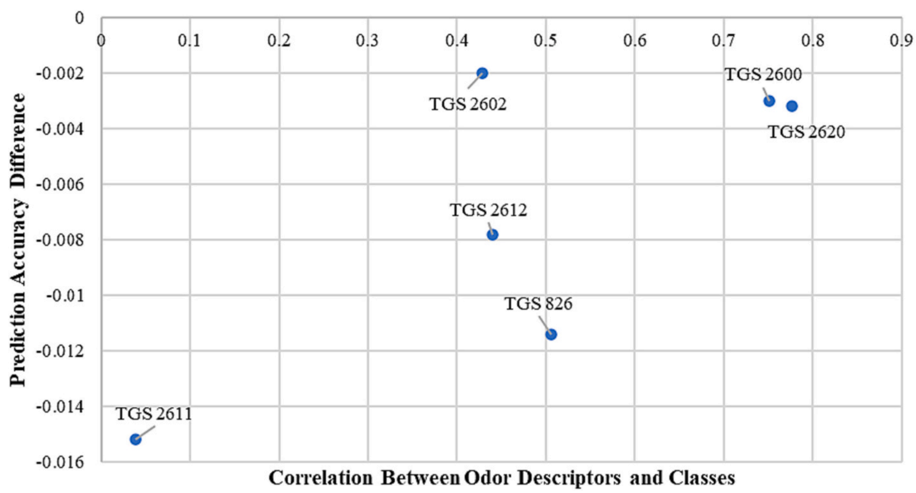

Fig. 3. Impact of descriptor correlation on prediction results.

coefficient ratings, TGS 2602 and TGS 826 have quite different personalities. This occurrence happens because, despite having identical information, sensor TGS 2602's information may be quieted by other sensors, but sensor TGS 826's information may be less quiet.

All sensors have the necessary knowledge regarding coffee classes, just like our intuitions do. We employ all sensors as part of our features, including the TGS 2602, TGS 2611, TGS 2612, TGS 826, TGS 2600, and TGS 2620.

3.2. Deep learning analysis

Cross-validation was used to compare Deep Learning accuracy with conventional techniques and represent the true state of machine learning. The other nine groups functioned as training sets when one group was used for testing. The ten-classification accuracy means were applied as a consequence. Because the leave-one-out cross-validation took longer and produced results that were identical to those of the 10-fold cross-validation, we opted for the former in this investigation. Cross-validation is the method we use to prevent overfitting. Node size and hidden layer size are covered in this section. The impact of the number of hidden layers on model performance is seen in Fig. 4. We used the first through fourth layers. From layer 1 to layer 3, we found that model performance improved. Layer three and the layer preceding it experienced an increase in model generalization, which is what caused this. Layer 4 is nonetheless less accurate than layer 3 because it is too complex for our data. A neural network with three principal hidden network (DHBP) layers is the last one we have.

Fig. 5 shows that the time it takes to process data goes down as the

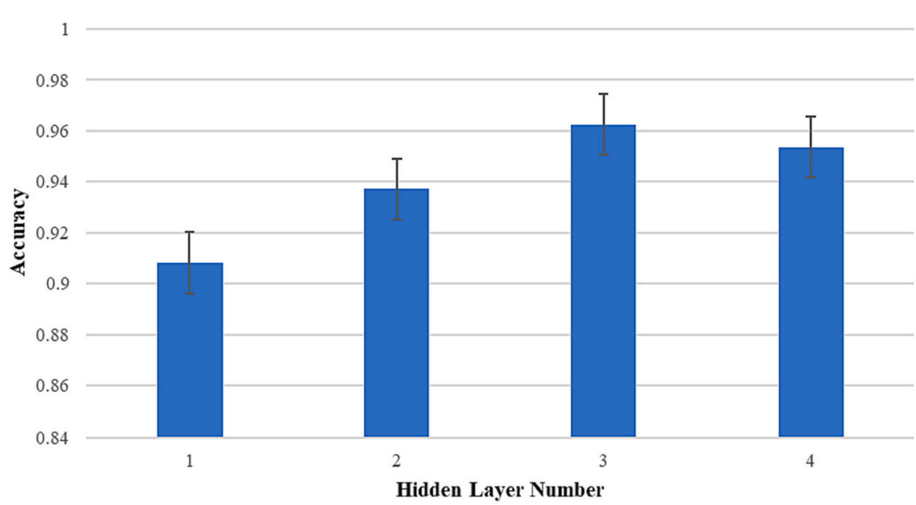

Fig. 4. Influence hidden layer number to accuracy.

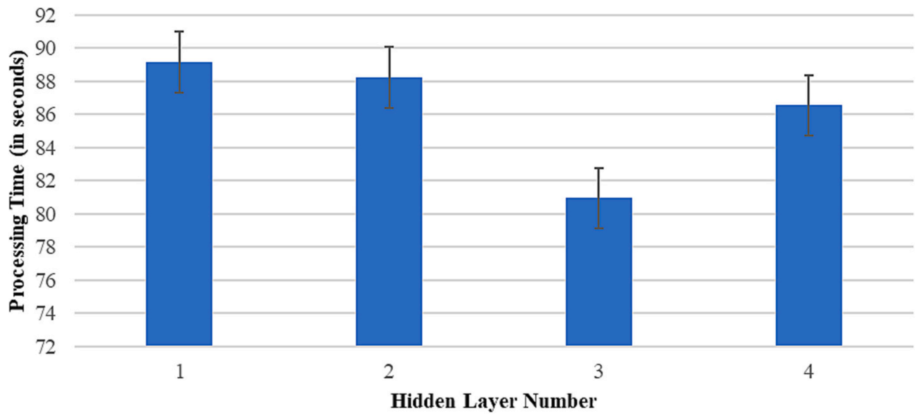

Fig. 5. Influence hidden layer to training time.

model gets closer to convergence. This happens because the appropriate model only needs to adjust the weight model a few times, whereas the inappropriate model requires more time to perform backpropagation.

This study looks at how likely it is that each layer has the same number of nodes as shown in Fig. 6. Using 10 to 150 nodes, we assess the model's performance. The performance of the system is linearly influenced by the number of nodes in a layer, from node 10 to node 60. The accuracy then declines until it reaches 70. This performance degradation is the result of a time-consuming computing process that was unable to thoroughly evaluate the data. Performance thus improves from nodes 81 to 91. At this time, the model has learned everything it can. The model finally reaches the maximum limit of model learning between 91 and 150 nodes, but the slow computational process makes model learning unsatisfactory. As a result, the parameter was set to be 91.

The effect of repetition times on accuracy is depicted in Fig. 7. Each iteration of our DHBP has comparatively higher accuracy. When the parameters are merely set in the second iteration, the accuracy decreases.

Other parameter settings used in this study are listed in Table 2. In order to make this paper more reproducible, we added a parameter table. We make use of a number of parameters, including optimizer, epsilon, and learning rate, that are frequently utilized in deep learning research.

Table 3 presents the classification results achieved through deep learning on cross-validation.

Predicting odor descriptor scores using electronic nasal signal data is an important research topic in coffee classification. Deep learning has shown superior performance in computer vision, speech recognition, and natural language processing compared to conventional machine learning techniques. Table 4 shows the results of the TGS gas sensor repeatability test. Based on the repeatability test results for each sensor, it shows that the TGS 2612 and TGS 2611 sensors are odor sensors that have good validity at various roasting temperatures. This discovery could be the answer to a managerial decision to choose a cheap but powerful odor sensor. This research found that the right number of hidden layers affects processing time in deep learning. Technical insights can be used in other domains.

Triple layers are used to overcome coffee prediction problem using *E*-Nose sensors using Pearson correlation analysis and odor descriptors. We should relax the requirement that all combinations contain the same number of descriptors and allow combinations to choose descriptors adaptively to get better combination effects. In the future, we will explore other odor sensors. We also want to explore other types of sensors to improve machine receptors such as tongue sensors.

3.3. Comparison performance across methods

According to our review, the classification of smelly coffee is closely related to two techniques that are often used. The first approach employs Linear Discriminant Analysis (LDA) to assess corpora [14,26]. A normal distribution is a crucial premise for LDA. Only a large enough dataset will allow for the normal distribution to be reached. The second technique combines Support Vector Machine (SVM) with Principal Component Analysis (PCA) [21]. Although PCA is effective at handling generalizations of data, it may actually result in information loss. As a result, we used three hidden layers to improve data generalization and a backpropagation baseline to lessen the dependence on the normal distribution. Table 3 displays our results in terms of accuracy.

In addition, we use two methods that are common in odor categorization to address a variety of issues. A decision tree is the first technique [28]. When the data set is small, the decision tree or discriminant classification method may become unstable. The first backpropagation method is used in the second method [29]. Backpropagation is effective in all domains, but can be improved by adding a hidden layer to capture the essence of the smell descriptor.

Backpropagation and SVM have a 0.1 accuracy difference, but DHBP

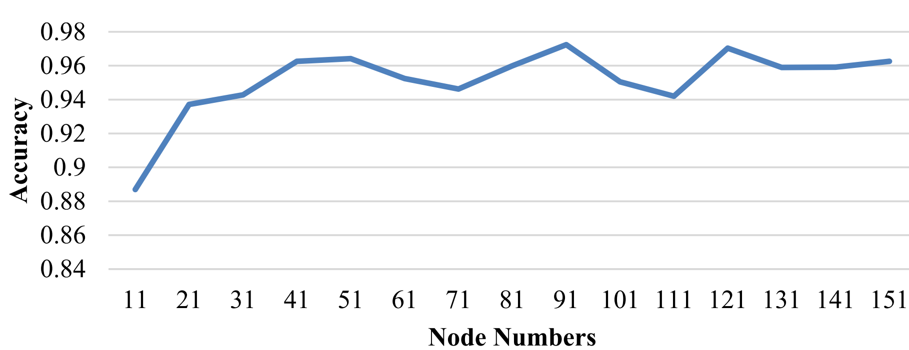

Fig. 6. Influence of node number to accuracy.

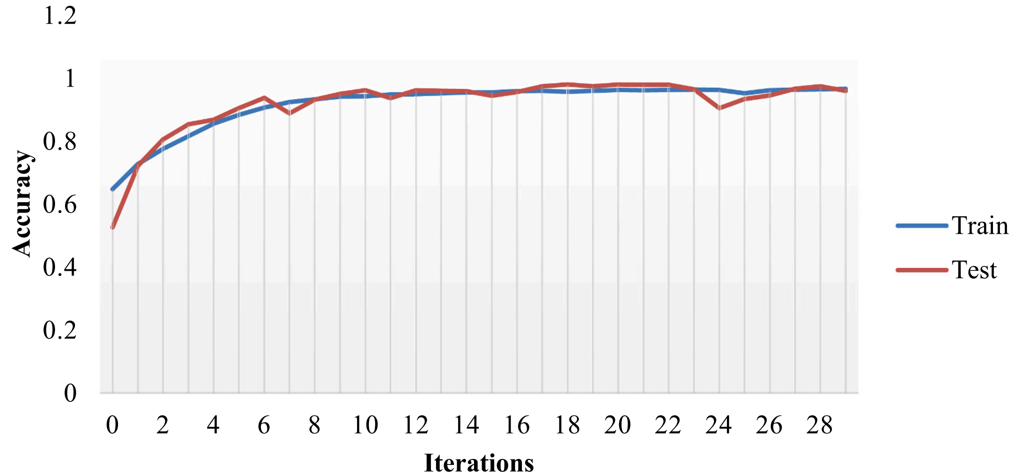

Fig. 7. Influence of iteration to accuracy.

Table 2
Parameters setting up of deep learning analysis.

Notation	Value	Definition
Optimizer	Adam	stochastic gradient descent method based on adaptive estimate of first order and second-order moments.
Epsilon	1e-07	a small positive constant that is typically used to avoid numerical instability and divide-by-zero errors during gradients of a loss function calculation
Learning Rate	0.001	a training parameter used to determine the weight correction value throughout the training process

still achieves higher accuracy. The developed model is applied to forecasting. This model outperforms all other machine learning techniques with a score of 0.937. We can infer from this outcome that the deep learning approach outperformed the machine learning approach. Our deep learning model outperforms the competition in terms of total accuracy, coming in at 0.97 with a 0.4 accuracy difference. Table 5 shows average number of each origin class and predicted class of cross-validation. From the table, we can notice that the most difficult profile is dark roast. The accuracy average of deep learning is 95%.

Deep learning predicts light roast, medium roast, and dark roast, with no miss classification. The system also gets a perfect classification when it comes to differentiates between light to medium roast and dark roast. The same thing also recorded in classifying the dark roast and light to medium. Therefore, the highest accuracy for overall is achieved when detecting the medium roast, which is 0.98.

4. Conclusions

This research successfully used deep learning and an electronic nose with TGS gas sensors to categorize Arabica coffee scents according to roasting temperature. The results of the repeatability test showed that the TGS sensor used had an STD below 20% which indicated that the sensor had good validity. TGS 2612 and TGS 2611 sensors are odor sensors that have good validity with an STD below 10% at various

Table 4
The result of TGS gas sensor repeatability test.

Sensor	Roasting temperature				
	185 °C	195 °C	205 °C	215 °C	225 °C
	STD (%)				
TGS 2600	23.4171	8.4409621	16.1116	16.1211121	17.0965
TGS 2602	13.8624	13.6001783	31.711	45.2457842	51.5936
TGS 2611	5.2252	6.0196961	10.0623	8.9897177	9.658
TGS 2620	19.3861	9.0462564	17.9403	17.9323349	18.912
TGS 2612	5.1299	3.2603214	3.1297	2.7724706	3.9926
TGS 826	18.3314	18.415294	36.6359	45.6665706	48.6132

Table 5
Comparison results of odor classifications.

Author	Year	Method	Accuracy	Std. Dev.
Machine Learning				
[9]	2011	Support Vector Machine	0.906	0.03
[30]	2017	Random Forest	0.639	0.00
[18,26]	2018 and 2019	Linear Discriminant Analysis	0.706	0.00
[27]	2020	Decision Tree	0.553	0.00
[21]	2021	Random Forest + Support Vector Machine	0.986	0.01
[6]	2022	Principle Component Analysis +Deep Neural Network	0.987	0.01
Deep Learning				
[28]	2017	1-Layer Hidden Backpropagation	0.937	0.02
[31]	2022	Principle Component Analysis +Support Vector Machine	0.985	0.01
[32]	2023	Artificial neural networks	0.903	0.01
This study	2023	Deep Learning	0.978	0.01

Table 3
Classification results by deep learning on cross-validation.

Odor data	Predicted group membership					Accuracy
	Light Roast	Light to Medium Roast	Medium Roast	Medium to Dark Roast	Dark Roast	
Light Roast	18,790.00	314.70	3.20	0.20	15.90	0.982
Light to Medium Roast	249.70	18,821.30	46.30	4.70	0.00	0.984
Medium Roast	2.00	20.20	18,917.30	104.50	88.00	0.988
Medium to Dark Roast	0.00	7.80	112.00	19,085.00	303.20	0.978
Dark Roast	6.20	0.00	160.10	616.00	18,333.70	0.959

roasting temperatures. Classification results by deep learning on cross-validation show good accuracy values, namely 98.2% for light roasts, 98.4% for light to medium roasts, 98.8% for medium roasts, 97.8% for Medium to dark roasts and 95.9% for dark roasts. With its objective and effective replacement for conventional approaches, this technology has potential uses for precise quality control and roasting level measurement in the coffee industry. The results of this study indicate that *E*-nose based on the TGS gas sensor array with deep learning analysis is able to detect and classify coffee types based on roasting time with good accuracy.

Author statement

We the undersigned declare that this manuscript is original, has not been published before and is not currently being considered for publication elsewhere. We confirm that the manuscript has been read and approved by all named authors and that there are no other persons who satisfied the criteria for authorship but are not listed. We further confirm that the order of authors listed in the manuscript has been approved by all of us.

CRediT authorship contribution statement

Suryani Dyah Astuti: Writing – review & editing, Visualization, Validation, Supervision, Funding acquisition, Data curation, Conceptualization. **Ihsan Rafie Wicaksono:** Writing – original draft, Methodology, Investigation, Formal analysis, Data curation, Conceptualization. **Soegianto Soelistiono:** Writing – original draft, Visualization, Validation, Methodology, Data curation, Conceptualization. **Perwira Annissa Dyah Permatasari:** Visualization, Resources, Project administration, Methodology, Formal analysis, Conceptualization. **Ahmad Khalil Yaqubi:** Writing – review & editing, Visualization, Validation, Software, Resources, Project administration, Formal analysis, Data curation, Conceptualization. **Yunus Susilo:** Visualization, Validation, Resources, Project administration, Methodology, Data curation, Conceptualization. **Cendra Devayana Putra:** Visualization, Validation, Project administration, Data curation, Conceptualization. **Ardiyansyah Syahrom:** Visualization, Validation, Methodology, Data curation, Conceptualization.

Declaration of competing interest

The authors state that they do not have any competing interests.

Data availability

Data will be made available on request.

References

[1] ICO, I.C.O, Exports by exporting countries to all destinations, Mon. Trade Stat. 1 (2022).
[2] M.A. Johnson, C.P. Ruiz-Diaz, N.C. Manoukis, J.C. Verle Rodrigues, Coffee berry borer (Hypothenemus hampei), a global pest of coffee: perspectives from historical and recent invasions, and future priorities, Insects. 11 (2010) 882.
[3] W.B. Sunarharum, D.J. Williams, H.E. Smyth, Complexity of coffee flavor: A compositional and sensory perspective, Int. Food Res. J. 62 (2014) 315–325.
[4] N. Bhumiratana, K. Chambers Adhikari, IV, E., Evolution of sensory aroma attributes from coffee beans to brewed coffee, LWT Food Sci. Technol. 44 (2011) 2185–2192.
[5] A. Huang, Y. Chao, E. de la Mora Velasco, A. Bilgihan, W. Wei, When artificial intelligence meets the hospitality and tourism industry: an assessment framework to inform theory and management, J. Hosp. Tour. 5 (2022) 1080–1100.
[6] A.I.F. Al Isyrofie, M. Kashif, A.K. Aji, N. Aidatuzzahro, A. Rahmatilah, Y. Susilo, S. D. Astuti, Odor clustering using a gas sensor array system of chicken meat based on temperature variations and storage time, Sens. BioSensing Res. 37 (2022) 100508.
[7] A.A.S. Pradhana, S.D. Astuti, M. Khasanah, R.K.D. Ardianti, Detection of gas concentrations based on age on *Staphylococcus aureus* biofilms with gas array sensors, in: AIP Conference Proceedings vol. 2314, AIP Publishing, 2020.
[8] S.D. Astuti, Y. Mukhammad, S.A.J. Duli, A.P. Putra, E.M. Setiawatie, K. Triyana, Gas sensor array system properties for detecting bacterial biofilms, J. Med. Signals Sens. 9 (2019) 158.
[9] E. Phaisangittisagul, H.T. Nagle, Predicting odor mixture's responses on machine olfaction sensors, Sensors Actuators B Chem. 155 (2011) 473–482.
[10] A.I.F. Isyrofie, A. Afifudin, R. Susilo, Y. Kholimatussa'diyah, S. Winarno, S. D. Astuti, Role of bacterial types and odor for early detection accuracy of bacteria with gas array, in: AIP Conf Proc 2554, 2023.
[11] S.N. Hidayat, K. Triyana, I. Fauzan, T. Julian, D. Lelono, Y. Yusuf, A.M. Peres, The electronic nose coupled with chemometric tools for discriminating the quality of black tea samples in situ, Chemosensors. 7 (2019) 29.
[12] J. Guo, Y. Cheng, D. Luo, K.Y. Wong, K. Hung, X. Li, ODRP: A deep learning framework for odor descriptor rating prediction using electronic nose, IEEE Sensors J. 21 (2021) 15012–15021.
[13] Y.H. Yu Wu, D.G. Li, H.C. Feng, Electrospun nanofibers for fast dissolution of naproxen prepared using a coaxial process with ethanol as a shell fluid, Appl. Mech. Mater. 662 (2014) 29–32.
[14] A. Sanaeifar, S. Mohtasebi, M. Ghasemi-Varnamkhasti, H. Ahmadi, J.S. Lozano Rogado, Development and Application of a New Low-Cost Electronic Nose for the Ripeness Monitoring of Banana Using Computational Techniques (PCA, LDA, SIMCA, and SVM), 2014.
[15] S. Buratti, N. Sinelli, E. Bertone, A. Venturello, E. Casiraghi, F. Geobaldo, Discrimination between washed Arabica, natural Arabica and Robusta coffees by using near infrared spectroscopy, electronic nose and electronic tongue analysis, J. Sci. Food Agric. 95 (2015) 2192–2200.
[16] K. Brudzewski, S. Osowski, A. Dwulit, Recognition of coffee using differential electronic nose, IEEE Trans. Instrum. Meas. 61 (2012) 1803–1810.
[17] W. Dong, J. Zhao, R. Hu, Y. Dong, L. Tan, Differentiation of Chinese Robusta coffees according to species, using a combined electronic nose and tongue, with the aid of chemometrics, Food Chem. 229 (2017) 743–751.
[18] S.Y. Kim, B.S. Kang, A colorimetric sensor array-based classification of coffees, Sensors Actuators B Chem. 275 (2018) 277–283.
[19] S. Romani, C. Cevoli, A. Fabbri, L. Dalla Alessandrini, M. Rosa, Evaluation of coffee roasting degree by using electronic nose and artificial neural network for off-line quality control, J. Food Sci. 77 (2018) (2012) C960–C965.
[20] S. Wang, K. Kirillova, X. Lehto, Travelers' food experience sharing on social network sites, J. Travel Tour. Mark. 34 (5) (2017) 680–693.
[21] S. Astuti, D. Tamimi, M.H. Pradhana, A.A. Alamsyah, K.A. Purnobasuki, H. Khasanah, M. Syahrom, Gas sensor array to classify the chicken meat with *E. coli* contaminant by using random forest and support vector machine, Biosens. Bioelectron.: X 9 (2021) 100083.
[22] D. Rabersyah, Identification of types of coffee grounds using electronic nose with backpropagation learning method, Aust. J. Electr. Electron. Eng. 5 (2016) 332–338.
[23] Y. Arimurti, K. Triyana, S. Anggrahini, Portable electronic nose as an instrument for discrimination of Java Robusta and Sumatran Robusta coffee aromas correlated with gas chromatography mass spectrometry, J. Phys. Sci. 10 (2018) 113–124.
[24] C.D. Putra, A.I.F. Al Isyrofie, S.D. Astuti, B.D. Putri, D.R. Ummah, M. Khasanah, A. Syahrom, Variational autoencoder analysis gas sensor array on the preservation process of contaminated mussel shells (*Mytilus edulis*), Sens. Bio-Sens. Res. 40 (2023) 100564.
[25] A.N. Gloess, A. Vietri, F. Wieland, S. Smrke, B. Schönbächler, J.A.S. López, C. Yeretzian, Evidence of different flavour formation dynamics by roasting coffee from different origins: on-line analysis with PTR-ToF-MS, Int. J. Mass Spectrom. 365 (2014) 324–333.
[26] F. Gottwalt, E. Chang, T. Dillon, CorrCorr: A feature selection method for multivariate correlation network anomaly detection techniques, Comput. Secur. J 83 (2019) 234–245.
[27] G.Y.F. Makimori, E. Bona, Commercial instant coffee classification using an electronic nose in tandem with the ComDim-LDA approach, Food Anal. Methods 12 (2019) 1067–1076.
[28] S. Wakhid, R. Sarno, S.I. Sabilla, D.B. Maghfira, Detection and classification of Indonesian civet and non-civet coffee based on statistical analysis comparison using E-nose, Int. J. Intell. Eng. Syst 13 (2020) 56–65.
[29] W. Zhao, Q.H. Meng, M. Zeng, P.F. Qi, Stacked sparse auto-encoders (SSAE) based electronic nose for Chinese liquors classification, Sensors 17 (2017) 2855.
[30] A. Chemura, O. Mutanga, Developing detailed age-specific thematic maps for coffee (*Coffea Arabica* L.) in heterogeneous agricultural landscapes using random forests applied on landsat 8 multispectral sensor, Geocarto Int. 32 (2017) 759–776.
[31] S.D. Astuti, A.I.F. Al Isyrofie, R. Nashichah, M. Kashif, T. Mujiwati, Y. Susilo, A. Syahrom, Gas array sensors based on electronic nose for detection of tuna (*Euthynnus Affinis*) contaminated by *Pseudomonas aeruginosa*, J. Med. Signals Sens 12 (2022) 306.
[32] A.A.S. Pradhana, S.D. Astuti, P.A.D. Permatasari, R. Agustina, A.K. Yaqubi, H. Setyawati, C.D. Putra, Sensor Array system based on electronic nose to detect borax in meatballs with artificial neural network, J. Electr. Comput. Eng. 23 (2023) 10.